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In situ synthesis of titanium doped hybrid metal–organic framework UiO-66 with enhanced adsorption capacity for organic dyes†

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Titanium in different amounts has been successfully doped into the zirconium-based metal–organic framework UiO-66 via an *in situ* synthesis method, resulting in a series of hybrid UiO-66-*n*Ti MOFs. These materials maintain a relatively high crystallinity and excellent structural stability. The addition of titanium has a significant effect on the crystal size and morphology of UiO-66. The UiO-66-*n*Ti MOFs exhibit a sphere-like crystal morphology with a smaller crystal size and a rougher surface compared to the octahedral UiO-66 crystals. The framework order and porosity of the UiO-66-*n*Ti MOFs decrease slightly due to titanium doping. The UiO-66-*n*Ti MOFs were studied as adsorbents for the removal of an organic dye from water. The results demonstrate that these hybrid materials have enhanced adsorption capacity for the organic dye Congo red in comparison with the parent UiO-66. UiO-66-2.7Ti with 2.7% titanium doping shows the highest adsorption capacity of 979 mg g^{−1}, which is three times higher than that of the parent UiO-66. The strong electrostatic attraction between the positively charged surface of UiO-66-2.7Ti and the negatively charged Congo red molecules was identified as the main driving force for the high adsorption capacity.

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1. Introduction

Water pollution has become one of the most serious environmental issues due to rapid industrialization and the steady increase in world population. Dye pollution is a common hazardous organic pollution in a variety of industrial wastewaters, for example, from the textile, leather, plastic, printing, cosmetics, pharmaceutical and food processing industries.^{1,2} Most dyes are toxic and even carcinogenic to human beings.³ Thus the efficient purification of wastewater by the removal or full degradation of dye pollutants is critically important.

Several approaches such as oxidation, chemical precipitation, photodegradation and membrane filtration have been applied to solve dye pollution issues.^{4,5} In view of the advantages of low cost and ease of operation, adsorption by porous materials has become one of the most promising approaches for dye removal during past decades.^{5,6}

Metal–organic frameworks (MOFs) are a unique class of porous crystalline materials that have been intensively studied over the past two decades due to their inherent design flexibility and their potential applications in gas adsorption/separation, luminescence and sensing, electrochemistry and catalysis, etc.^{7–18} MOFs are composed of desired metal clusters connected by organic linkers that are rich in functionalities, possessing ordered network structures with permanent porosity.^{19,20} Therefore, high surface area, devisable pore size and shape, adjustable functional groups and numerous active metal sites make MOFs promising candidates for the removal of dyes from wastewater. Recently, doping foreign metals in the MOF structure to synthesize mixed-metal hybrid MOFs has been shown to be an efficient strategy to improve the liquid- and gas-phase adsorption performance; the performance enhancement is primarily attributed to the increased surface area or the additional active sites.^{21–23} For example, alkali metal ions (Li⁺, Na⁺ and K⁺) were successfully doped in MOF-5 by an *in situ* crystallization method.²² The alkali metal-doped

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MOF-5 had a higher surface area than MOF-5, and the H₂ uptake capacity was also enhanced by the increased polarizability from the doping of alkali metals. A novel bimetallic MIL-101(Cr, Mg) MOF was obtained by doping Mg²⁺ in the synthesis of MIL-101(Cr).²⁴ The MIL-101(Cr, Mg) material exhibited a higher CO₂ adsorption capacity and CO₂/N₂ selectivity due to the increased surface area as well as the stronger interaction between MIL-101(Cr, Mg) and CO₂. Furthermore, partial doping of the Cu-BTC MOF with Ni²⁺ was successfully carried out by a solvothelmal method.²⁵ The synthesized Ni/Cu-BTC was applied to the adsorption of the organic dye from the aqueous solution, and showed superior adsorption capacity than the parent Cu-BTC due to the higher affinity between Ni²⁺ and the organic dye molecules. Besides, a new three-dimensional bimetallic MOF [CoNi(μ₃-tp)₂(μ₂-pyz)₂] (tp = terephthalic acid, pyz = pyrazine) was hydrothermally synthesized using O- and N-donor linkers.²⁶ For the removal of the organic dye Bismarck from the aqueous suspension, this bimetallic material exhibited good activity and stability for at least three cycles.

As one of the most important MOFs, Zr-BDC MOF, called UiO-66 (*i.e.*, Zr₆O₄(OH)₄(CO₂)₁₂), has been extensively studied due to its high porosity and excellent structural stability toward high temperature, mechanical pressure and chemical (acidic/basic) environment.^{27–29} Recently, Ce³⁺ modified UiO-66 has been successfully synthesized under solvothermal conditions. The doping of Ce³⁺ not only altered the surface chemistry and texture of UiO-66, but also provided new redox active sites, resulting in enhanced properties in NO₂ adsorption and improved catalytic performance in alcohol decomposition.^{21,30} Owing to the chemical similarity between Zr⁴⁺ and Ti⁴⁺, the incorporation of Ti⁴⁺ in UiO-66 may lead to an optimized performance in various applications. Cohen *et al.* first reported the successful synthesis of Ti⁴⁺ exchanged UiO-66(Zr/Ti) MOFs by the post-synthetic exchange (PSE) method.³¹ The smaller Ti atom incorporated in UiO-66 has been found to decrease the pore size of the octahedral cages within the framework and to increase the affinity of CO₂.^{32,33} Besides, the photocatalytic properties of UiO-66 were also improved by doping a suitable amount of Ti⁴⁺ in the framework.^{34,35}

Inspired by the studies mentioned above, we selected UiO-66 as the parent MOF and Ti⁴⁺ as the foreign doping metal to explore the possibility of the facile synthesis of Ti⁴⁺ doped hybrid UiO-66-*n*Ti MOFs via an *in situ* crystallization process. The performance of UiO-66-*n*Ti MOFs as adsorbents for organic dyes was evaluated to investigate the influence of Ti⁴⁺ doping. The synthesized materials were structurally characterized by XRD, ICP, SEM-TEM, FT-IR, Raman spectroscopy, XPS, TGA and argon sorption. Compared with the parent UiO-66, the UiO-66-*n*Ti MOFs show highly selective adsorption of anionic organic dye Congo red, the highest adsorption capacity is three times higher than that of the parent UiO-66. The Langmuir isotherm model and pseudo-second-order model were found to be suitable to describe the adsorption process. A possible adsorption mechanism is also discussed.

2. Experimental

2.1 Chemicals

Zirconium tetrachloride (ZrCl₄, Sigma-Aldrich, 98%), terephthalic acid (H₂BDC, 98%) and titanium isopropoxide (C₁₂H₂₈O₄Ti, TPOT) were purchased from Aladdin. *N,N*-Dimethylformamide (DMF), acetic acid (CH₃COOH) and absolute methanol (CH₃OH) were obtained from Shanghai Chemical Reagent Inc. of the Chinese Medicine Group. Different dyes, Congo red (CR), Methylene blue (MB), Methyl orange (MO) and Rhodamine B (RhB) were purchased from Tianjin Fuyu Fine Chemical Co. Ltd. Deionized water was produced in the Dalian University of Technology. All chemicals were utilized as received without further purification.

2.2 Synthesis of the samples

The synthesis of the parent UiO-66 was carried out by dissolving 1.362 mmol of ZrCl₄ (0.317 g) and H₂BDC (0.226 g) in 30 mL of DMF; 2.5 mL of acetic acid was then added to the solution. The obtained solution was loaded into a 100 mL Teflon-lined steel autoclave and then placed in a preheated oven at 120 °C for 24 h under static conditions. The UiO-66-*n*Ti MOFs (*n* represents the mass fraction of Ti⁴⁺) were synthesized by following the same procedure but with varying amounts of Ti⁴⁺ being added (titanium isopropoxide was used as the Ti⁴⁺ precursor). The selected molar ratios of Ti/Zr were 5%, 10%, 15%, 20% and 30%. After crystallization, the white products were collected by centrifugation, thoroughly washed with DMF and methanol, and then dried overnight under vacuum at 60 °C.

2.3 Characterization methods

Powder X-ray diffraction (XRD) patterns were recorded on a Rigaku SmartLab(9) diffractometer with Cu Kα X-rays at a scanning rate of 10° min⁻¹ between 5° and 50°. The tube voltage and current used were 40 kV and 100 mA, respectively. The concentrations of Ti⁴⁺ in the UiO-66-*n*Ti MOFs were determined by using an Inductively Coupled Plasma-Atomic Emission Spectrometer (Optima 2000DV, PerkinElmer, Waltham, MA, USA). Raman spectra were recorded using a DRX Raman Microscope (Thermo Fisher Scientific, USA) with the excitation wavelength of 532 nm. SEM images were recorded using a NOVA NanoSEM 450 field-emission scanning electron microscope (FEI, Hillsboro, Oregon, USA). Transmission electron microscopy (TEM) images were recorded using a Tecnai G2 20 S-twin transmission electron microscope (FEI, Hillsboro, OR, USA). FT-IR spectra were recorded using an EQUINOX55 Fourier transform infrared spectrometer (Bruker, Germany) by using the KBr pellet technique. X-ray photoelectron spectra (XPS) of all samples were recorded using a VG ESCALAB250 Electron Spectrometer with a monochromatic Al Kα (1486.6 eV) at 15 kV and 10 mA, and all binding energies were referenced to the C 1s peak (284.6 eV). Thermogravimetric analysis (TGA) was performed using an SDT Q600 thermogravimetric analyzer (TA Instruments, USA) in the temperature range of 25–800 °C under air flow at a

heating rate of 10 °C min⁻¹. Ar sorption isotherms were recorded at 87 K using a Quantachrom Autosorb-iQ instrument. Before the measurements, all samples were outgassed at 150 °C overnight. Zeta potentials of the samples in ultrapure water were measured using a zeta potential analyzer (ZETASIZER nanoseries Nano-ZS90, Malvern instruments).

2.4 Dye adsorption experiments

All adsorption experiments were carried out at 35 °C on a thermostated shaker in a dark environment. 25 mg of the adsorbents were mixed with 50 mL of dye aqueous solution in an Erlenmeyer flask. The concentration of each dye solution was adjusted from 100 to 1200 mg L⁻¹. After adsorption for different time periods (0–4 h), the adsorbents were separated from the stock solution through a syringe filter (PTFE, hydrophobic, 0.22 µm), the concentration of the clear solution was analyzed by using a UV-visible spectrophotometer at a λ_{max} corresponding to the maximum adsorption for the dye solution. The adsorption capacity Q_e (mg g⁻¹) of the dye was calculated using the following equation:

$$Q_e = \frac{(C_0 - C_e) \times V}{M}$$

where C_0 and C_e are the initial and equilibrium concentrations of the dye (mg L⁻¹), M is the mass of the adsorbent (g) and V is the volume of the solution (L). Prior to the adsorption experiments, the adsorbents were treated under vacuum at 150 °C for 3 h.

3. Results and dissolution

3.1 Characterization of the UiO-66-*n*Ti MOFs

The Ti⁴⁺ concentrations of the UiO-66-*n*Ti MOFs were determined by ICP measurements. As illustrated in Table 1, the concentration of Ti⁴⁺ increased gradually from around 0.7 to 4.0% with the increase in the initial doping amount of Ti⁴⁺ in the synthesis. Most real molar ratios of Ti/Zr were slightly higher than the initial values in the synthesis.

The XRD patterns of the samples are shown in Fig. 1. All the XRD patterns of the UiO-66-*n*Ti MOFs maintain the unique characteristic diffraction peaks of the pure phase UiO-66 with relatively high crystallinity. No extra peaks associated with Ti⁴⁺

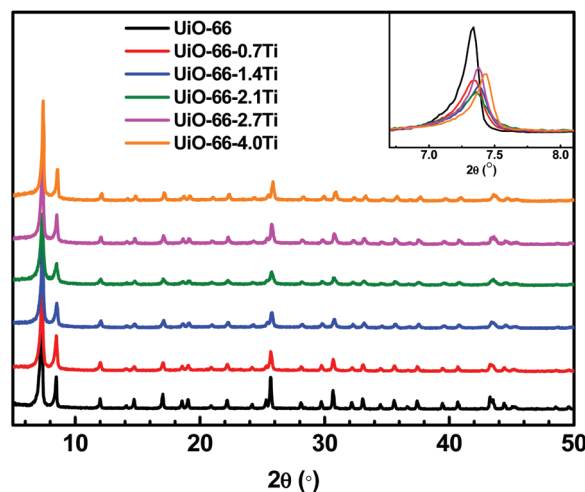


Fig. 1 XRD patterns of the synthesized samples.

oxides or other Ti-related impurities can be observed from the XRD patterns. The Raman spectra of the UiO-66-*n*Ti MOFs shown in Fig. S1† further indicate the pure phase of the samples.³⁶ However, compared to the parent UiO-66, the intensity of the diffraction peaks of the UiO-66-*n*Ti MOFs decreased slightly, and the peaks at around $2\theta = 7.3^\circ$ shifted to higher values (see the inset of Fig. 1). Similar phenomena have been observed in many transitional metal doping systems, which indicate the partial substitution of larger Zr⁴⁺ in the [Zr-O] clusters by smaller Ti⁴⁺, resulting in shrunk crystal lattice and suppression of crystal growth for the UiO-66-*n*Ti MOFs.^{34,37–41}

The SEM and EDS elemental maps of the samples are shown in Fig. 2. The parent UiO-66 shows the octahedral morphology with a mean crystal size of 265 nm. On comparing the SEM images of the UiO-66-*n*Ti MOFs with those of the parent UiO-66, significant differences can be seen. At a low Ti⁴⁺ concentration, the mean crystal size of UiO-66-0.7Ti decreased sharply from 265 to 65 nm along with the morphology change from octahedra to sphere-like crystals with rougher surfaces. It is noteworthy that when the Ti⁴⁺ concentration was further increased, the mean crystal size of the samples showed an increasing trend from 65 to 183 nm instead of further decreasing. Meanwhile, all the UiO-66-*n*Ti MOFs maintained the sphere-like morphology. The remarkable change in the crystal size and morphology of the UiO-66-*n*Ti MOFs further suggests that the Ti⁴⁺ doping has restrained the growth of UiO-66 crystals.^{37,39} In addition, the TEM images of UiO-66-2.7Ti shown in Fig. S2† reveal the homogeneous-phase crystal morphology. Therefore, the SEM and TEM images of the UiO-66-*n*Ti MOFs together demonstrate the homogeneous crystal morphology rather than a mixture of Ti-related impurity and individual UiO-66 crystals.^{42–44} The elemental maps of UiO-66-4.0Ti demonstrate the homogeneous distributions of Zr and Ti as well as C and O in the framework. The XPS spectra in the Ti 2p region of the UiO-66-*n*Ti MOFs further confirm the Ti⁴⁺ doping. As illustrated in Fig. 3, the Ti 2p_{3/2} peak at around 457.5 eV and the Ti 2p_{1/2} peak at around 463.2 eV appeared

Table 1 ICP data for the concentration of Ti⁴⁺ and the real molar ratio of Ti/Zr in the UiO-66-*n*Ti MOFs

Sample	ICP for the concentration of Ti ⁴⁺ (wt%)	ICP for the molar ratio of Ti/Zr (%)	Feeding molar ratio of Ti/Zr (%)
UiO-66-0.7Ti	0.7	4.4	5
UiO-66-1.4Ti	1.4	10.3	10
UiO-66-2.1Ti	2.1	18.6	15
UiO-66-2.7Ti	2.7	21.4	20
UiO-66-4.0Ti	4.0	33.2	30

Before the ICP measurements, all samples were heated at 150 °C for 3 h under vacuum.

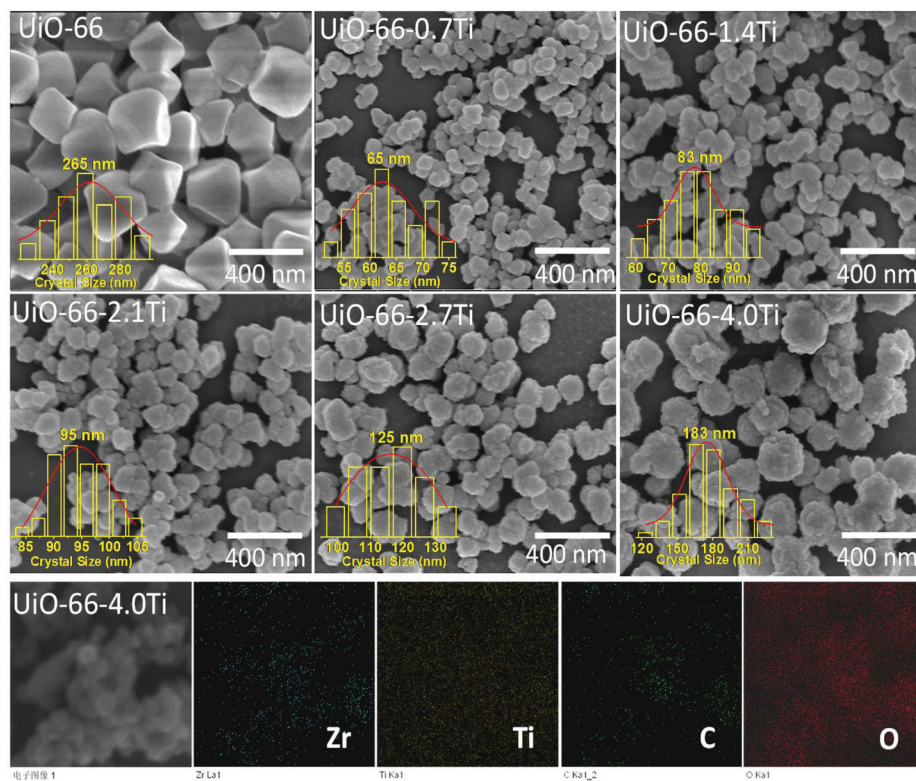


Fig. 2 SEM images of the synthesized samples and EDS elemental maps of UiO-66-4.0Ti.

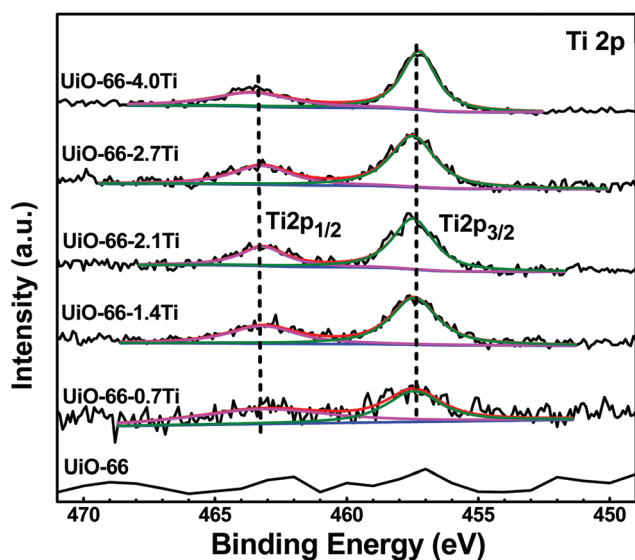


Fig. 3 Ti 2p XPS spectra of the synthesized samples.

for the UiO-66-*n*Ti MOFs corresponding to the octahedral and tetrahedral coordinated Ti^{4+} , respectively.^{45–47} As XPS is a highly surface-sensitive technique for the analysis of elements, the surface molar ratio of Ti/Zr obtained by XPS analysis provides some information about the material structures (The surface molar ratios of Ti/Zr in the UiO-66-*n*Ti MOFs are

presented in Table S1.†).³⁵ At a lower Ti^{4+} doping amount (for samples UiO-66-0.7Ti and UiO-66-1.4Ti), the surface molar ratio of Ti/Zr was higher than the bulk Ti/Zr molar ratio based on the ICP analysis shown in Table 1, suggesting higher Ti^{4+} on the surface. In contrast, samples UiO-66-2.1Ti, UiO-66-2.7Ti and UiO-66-4.0Ti with increased Ti^{4+} doping amounts showed a higher bulk molar ratio of Ti/Zr than the surface molar ratio of Ti/Zr, indicating that more Ti^{4+} were incorporated within the framework with the increase in the initial doping amount of Ti^{4+} in the synthesis.

Fig. 4 shows the FT-IR spectra of the synthesized samples. All samples display similar peaks in general. The small band around 1660 cm^{-1} is ascribed to the C=O asymmetric stretch of the residual DMF within the pores.⁴⁸ The two intense bands around 1590 cm^{-1} and 1401 cm^{-1} are associated with the OCO asymmetric and symmetric stretch vibrations in the carboxylate group in H_2BDC , respectively. The small band around 1510 cm^{-1} represents the vibration of C=C in the benzene ring.²⁸ At lower frequencies, the bands around 810 cm^{-1} , 748 cm^{-1} and 710 cm^{-1} are mixed with the C-H vibration, C=C stretch, OH bend and OCO bend in H_2BDC . The bands around 680 cm^{-1} and 480 cm^{-1} represent the Zr- μ_3 -O and Zr- μ_3 -OH stretch, respectively, and the band around 550 cm^{-1} belongs to the Zr-(OC) asymmetric stretch.^{28,49} Although the incorporation of Ti^{4+} into the UiO-66 frameworks cannot be clearly seen on the FT-IR spectrum, the Zr- μ_3 -O stretch in the [Zr-O] clusters at around 682 cm^{-1} in the parent UiO-66 showed a gradual red shift to around 664 cm^{-1} with the

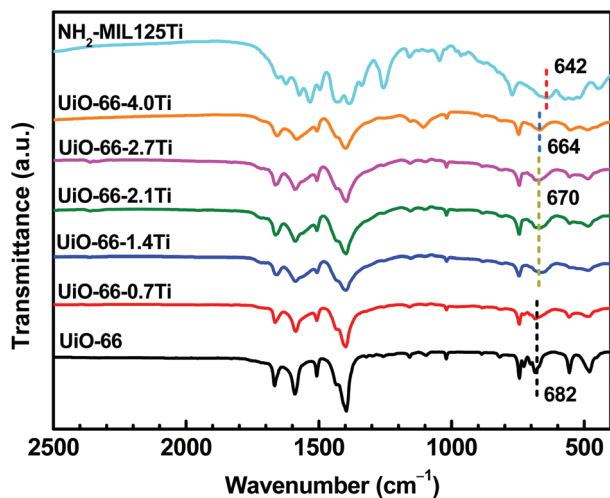


Fig. 4 FT-IR spectra of the UiO-66-*n*Ti MOFs and NH₂-MIL125(Ti).

increase in the Ti⁴⁺ concentration, implying the slight modification to the structure in the UiO-66-*n*Ti MOFs. Besides, as can be seen from Fig. 4, the Ti-BDC MOF NH₂-MIL-125(Ti) also possesses a band at around 642 cm⁻¹ which is assigned to the (O-Ti-O) vibrations.^{50–52} Thus the red shift of the Zr-μ₃-O stretch in the UiO-66-*n*Ti MOFs could be caused by the incorporation of Ti⁴⁺ in the [Zr-O] clusters. Therefore, the formation of the Ti⁴⁺ doped hybrid UiO-66-*n*Ti MOFs is confirmed by the results of XRD, EDS, XPS and FT-IR.

Due to the different coordination number of Ti⁴⁺ and Zr⁴⁺ (6-coordination for Ti⁴⁺ and 8-coordination for Zr⁴⁺), Ti⁴⁺ in the UiO-66 framework cannot bind to the same number of oxo groups, hydroxyl groups and the linkers as does Zr⁴⁺.⁵³ As a result, structural defects were created. The structural deficiency of UiO-66 is usually studied by TGA analysis.^{28,54–56} Fig. 5 shows the TGA-DTG curves of the synthesized samples. As can be seen, all samples show similar thermal stabilities by exhibiting three steps of weight loss, including the loss of

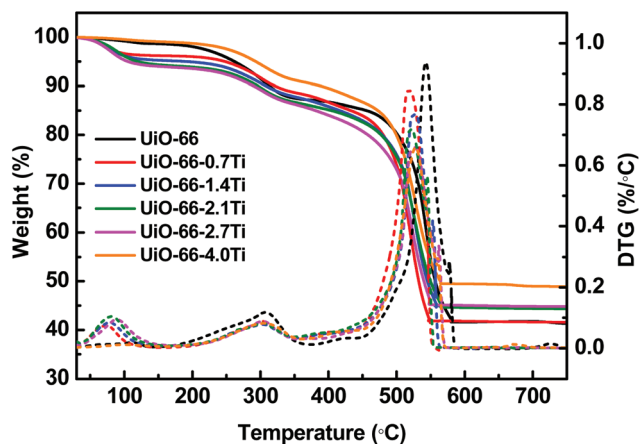


Fig. 5 TGA-DTG curves of the synthesized samples.

physisorbed water and residual solvent molecules before 200 °C; the dehydroxylation process in the 200–350 °C interval and the complete decomposition of the framework are caused by the combustion of the H₂BDC linkers after 450 °C. Therefore, the weight loss value of the samples in the 450–700 °C interval can reveal the percentage of H₂BDC in the structure. Referring to a chemical formula per Zr atom of ZrO(CO₂)₂(C₆H₄) of a defect-free dehydrated UiO-66, its weight loss in the 450–700 °C interval is around 54.6%.²⁸ However, all the synthesized samples have lower weight loss values below 44% in this interval (as shown in Table S2†), indicating the existence of the linker defects in the structure. Additionally, the weight loss values of the UiO-66-*n*Ti MOFs in the 450–700 °C interval are lower than those of the parent UiO-66, which confirms that the UiO-66-*n*Ti MOFs have a more defective structure than the parent UiO-66. The structural features of the UiO-66-*n*Ti MOFs on the TGA curves are in agreement with the XRD and SEM results in that the doping of Ti⁴⁺ has restrained the construction of the ordered framework structure of UiO-66 and led to more structural defects.

The porosities of the samples were evaluated by argon sorption measurements at 87 K. The sorption isotherms and pore size distributions are shown in Fig. 6. All samples exhibited a type I sorption isotherm, indicating a typical microporous structure. However, the argon uptake capacities (thus the porosities) of the samples varied depending on the concentration of the incorporated Ti⁴⁺. As can be seen from Fig. 6a, the porosity increased first (for UiO-66-0.7Ti) and was followed by a systematical decrease with the increase in the Ti⁴⁺ concentration. Additionally, as illustrated in Fig. 6b, the parent UiO-66 is dominated by a large amount of micropores between 6 and 10 Å, with only a very small fraction of pores between 15 and 20 Å. However, the amount of micropores decreased significantly after the Ti⁴⁺ incorporation. The porous structure data are shown in Table 2. The porosity of the parent UiO-66 (1358 m² g⁻¹, 0.48 cm³ g⁻¹ and 0.54 cm³ g⁻¹ for the BET surface area, micropore volume and total pore volume, respectively) is within the typical range reported in the literature.⁵⁷ Compared to the parent UiO-66, increases in both the BET surface area and micropore volume were observed for UiO-66-0.7Ti at a low Ti⁴⁺ concentration. Since UiO-66-0.7Ti has the smallest crystal size of about 65 nm among all the samples (Fig. 2), the increase of the porosity is presumably attributed to the formation of interparticle porosity.^{58,59} However, both the BET surface area and the micropore volume decreased systematically (from 1444 to 792 m² g⁻¹ for the BET surface area and from 0.52 to 0.26 cm³ g⁻¹ for the micropore volume) for the samples with further increased Ti⁴⁺ concentrations. The decrease in the BET surface area and micropore volume of the UiO-66-*n*Ti MOFs could be attributed to the loss of the well-ordered structure due to Ti⁴⁺ incorporation, as confirmed by the XRD and TGA results.⁶⁰ A similar phenomenon was also observed by Liu *et al.*³⁵ Furthermore, the smaller crystal sizes of the UiO-66-*n*Ti MOFs compared to the parent UiO-66 (as illustrated in Fig. 2) may also cause the slight decrease of the micropore porosity.⁶¹

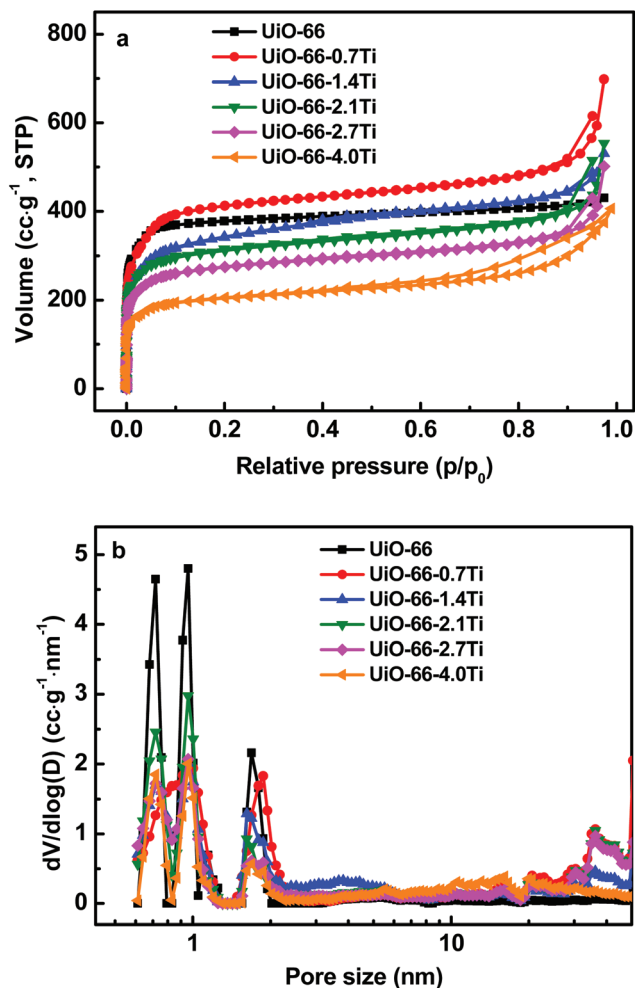


Fig. 6 (a) Argon sorption isotherms and (b) pore size distributions of the synthesized samples. The pore size distributions are measured in the 0–50 nm range.

Table 2 Textural structure data of the synthesized samples

Sample	S_{BET} ($\text{m}^2 \text{g}^{-1}$)	V_{micro} ($\text{cm}^3 \text{g}^{-1}$)	V_{total} ($\text{cm}^3 \text{g}^{-1}$)
UiO-66	1358	0.48	0.54
UiO-66-0.7Ti	1444	0.52	0.72
UiO-66-1.4Ti	1144	0.43	0.60
UiO-66-2.1Ti	1056	0.39	0.50
UiO-66-2.7Ti	929	0.35	0.66
UiO-66-4.0Ti	792	0.26	0.44

3.2 Dyes adsorption on the UiO-66-*n*Ti MOFs

3.2.1 Adsorption selectivity. The adsorption selectivity of the UiO-66 with and without Ti^{4+} doping was first investigated. Four common organic dyes including cationic dyes Methylene Blue (MB) and Rhodamine B (RhB) and anionic dyes Congo Red (CR) and Methyl Orange (MO) were applied as the adsorbates (the molecular structures of the dyes and their estimated molecular sizes are listed in Table S3†). The UV-vis adsorption spectra of the dyes before and after the adsorption treatment

with the parent UiO-66 and UiO-66-2.7Ti are displayed in Fig. 7. As can be seen, UiO-66-2.7Ti shows the obvious adsorption selectivity for CR. The removal rate of CR reaches 72% after 4 h adsorption, which is only 24% by the parent UiO-66. For MB, MO and RhB, UiO-66-2.7Ti shows very close adsorption abilities with the removal rates of 29%, 28% and 26%, respectively. These results indicate that UiO-66-2.7Ti can effectively remove CR from water. However, on the parent UiO-66, the adsorption abilities of the four dyes are similar and are at a relatively low level. After 4 h of adsorption, the removal rates are 36%, 27% and 11% for MB, MO and RhB, respectively. Therefore, CR was selected as the adsorbate to further investigate the adsorption capacity of the UiO-66-*n*Ti MOFs.

3.2.2 Adsorption kinetics. For dye adsorption, it is important to study the adsorption kinetics as it provides information about the mechanism of adsorption.⁶² In this work, the adsorption kinetics was studied on the parent UiO-66 and UiO-66-2.7Ti with an initial CR concentration of 400 mg L^{-1} . Fig. 8 shows the time-resolved adsorption capacity on the two samples. As illustrated, the adsorption amount showed a rapid increase with prolonging contact time. The adsorption equilibrium can almost be reached within 60 min. The adsorption capacity of UiO-66-2.7Ti reaches 607 mg g^{-1} , much higher (251 mg g^{-1}) than that of the parent UiO-66. The adsorption data were studied using pseudo-first-order and pseudo-second-order equations (eqn (1) and (2) in the ESI†) to investigate the adsorption kinetic characteristics. The results (Fig. S3 and Table S4†) show that the adsorption data fit well with the pseudo-second-order model, with R^2 (the correlation coefficient) being higher than 0.99, suggesting that the adsorption process of CR seems to be controlled by the chemical process.⁶³

3.2.3 Adsorption isotherms. Subsequently, in order to further investigate the adsorption ability of the UiO-66-*n*Ti MOFs, the adsorption isotherms for CR were obtained at 35°C after adsorption for 4 h with different initial CR concentrations ($100\text{--}1200 \text{ mg L}^{-1}$). The results are shown in Fig. 9. As can be seen, the parent UiO-66 and UiO-66-0.7Ti show a relatively low adsorption capacity of about 280 mg g^{-1} . With the increase in the Ti^{4+} concentration in the UiO-66 framework, the adsorption capacity has been significantly increased. UiO-66-2.7Ti shows the highest adsorption capacity of 979 mg g^{-1} , followed by UiO-66-2.1Ti at 869 mg g^{-1} , whereas the adsorption capacity of UiO-66-4Ti decreases to 608 mg g^{-1} . The adsorption ability of UiO-66-2.7Ti on CR is comparable to or higher than those of some MOFs adsorbents listed in Table 3. Furthermore, the equations of Langmuir and Freundlich (eqn (3) and (4) in the ESI†) were applied for describing the adsorption on the parent UiO-66, UiO-66-1.4Ti and UiO-66-2.7Ti. The results are shown in Fig. S4 and Table S5.† In comparison with the Freundlich equation, the Langmuir equation is more suitable for fitting the adsorption data, suggesting the homogeneous distribution of surface active sites of the samples, with the adsorption limited to monolayer coverage.⁶⁴

The effect of the synthesis method for the incorporation of the Ti^{4+} was also investigated. We prepared sample UiO-66(Ti)

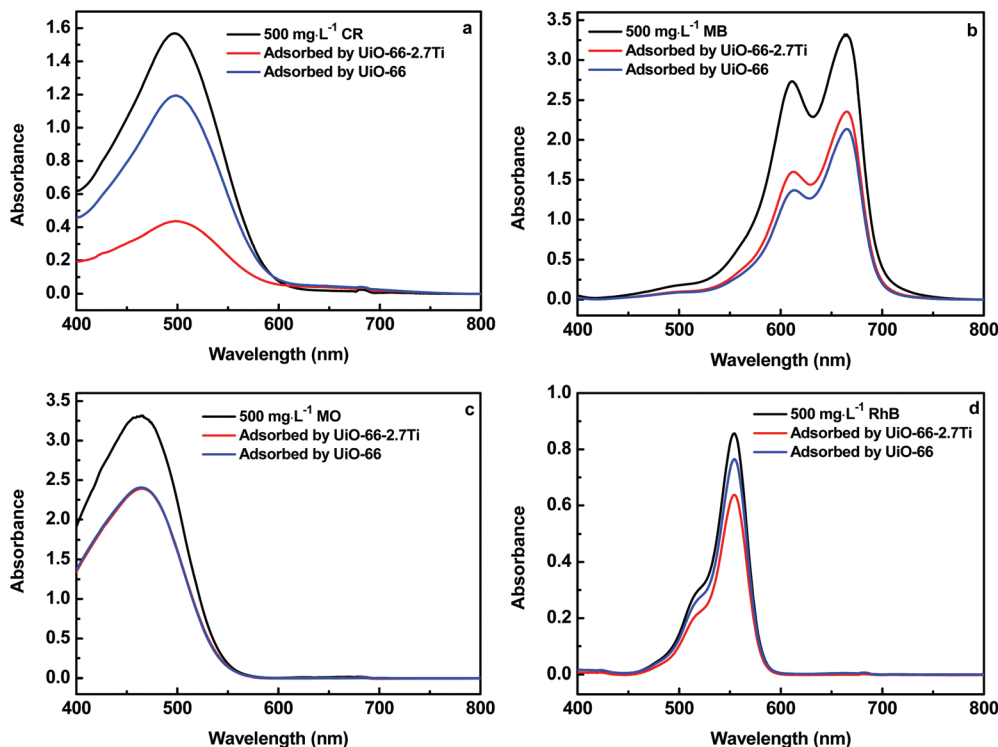


Fig. 7 UV-vis adsorption spectra of the organic dyes before (black curve) and after (red and blue curves) adsorption treatment with UiO-66-2.7Ti and the parent UiO-66 ($t = 4$ h, $\text{pH} = 7$, dyes = 500 mg L^{-1} , does = 0.5 g L^{-1} , $T = 35^\circ\text{C}$).

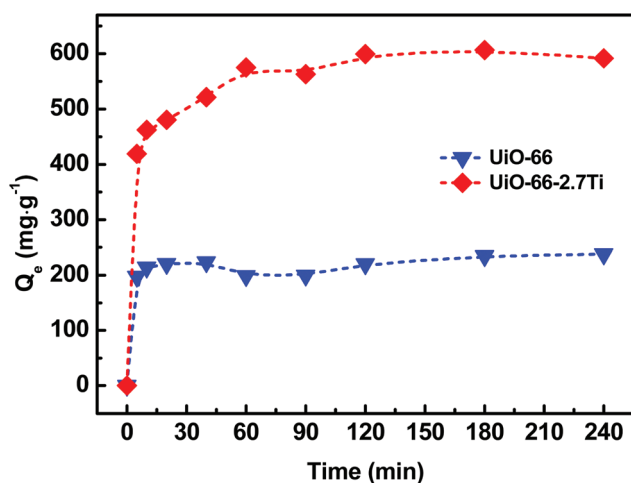


Fig. 8 Time-resolved adsorption capacity of CR on the parent UiO-66 and UiO-66-2.7Ti ($t = 4$ h, $\text{pH} = 7$, CR = 400 mg L^{-1} , does = 0.5 g L^{-1} , $T = 35^\circ\text{C}$).

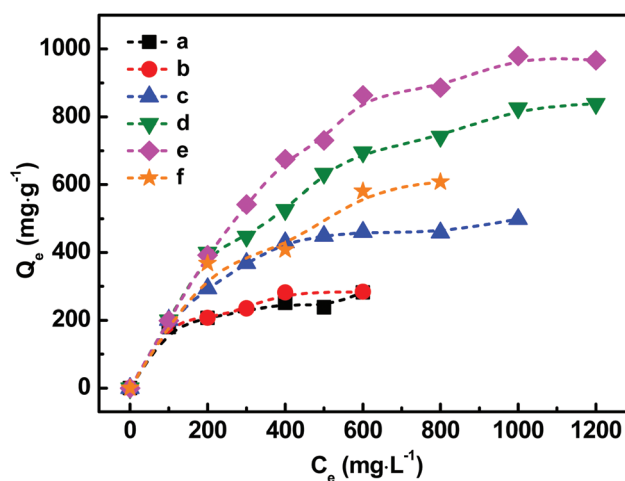


Fig. 9 Adsorption isotherms of CR on the parent UiO-66 (a), UiO-66-0.7Ti (b), UiO-66-1.4Ti (c), UiO-66-2.1Ti (d), UiO-66-2.7Ti (e) and UiO-66-4Ti (f) ($t = 4$ h, $\text{pH} = 7$, does = 0.5 g L^{-1} , $T = 35^\circ\text{C}$).

with the Ti/Zr feeding molar ratio of 20% (ICP analysis showed that the Ti^{4+} concentration was around 2.5%) *via* a modified post-grafting method³⁵ (the synthesis details are shown in the ESI†). The adsorption capacity of CR on UiO-66(Ti) was investigated with an initial CR concentration of 1000 mg L^{-1} , other conditions remained the same. However, UiO-66(Ti) shows a considerably low removal rate with adsorption capacity of only about 100 mg g^{-1} (Fig. S5†). Therefore, this result further

demonstrates that UiO-66-2.7Ti synthesized through the *in situ* method is the most promising adsorbent for CR in this work.

3.2.4 Possible adsorption mechanism. The adsorptive removal of organic dyes by MOF adsorbents in the liquid phase is a complicated process, usually influenced by factors such as electrostatic attractions, pore/size-selective adsorption, hydrogen bonding, acid-base interactions and π - π interactions/stacking.^{5,6} In this work, although a more detailed

Table 3 Comparison of CR adsorption capacities in this work with various MOFs adsorbents

Adsorbents	Q_e (mg g ⁻¹)	Ref.
MIL-68(In) nanorods	1204	65
50Ni/Cu-BTC	1078	25
UiO-66-2.7Ti	979	This work
Cu-BTC	877	66
UiO-66-2.1Ti	869	This work
{[Cu ₃ (btb) ₃ (nbta) ₂](H ₂ O) ₂] _n }	656	67
UiO-66	283	This work
[Cd ₂ (oba) ₂ (4-bpdb) ₂] _n (DMF) _x	113	68
[Zn(BDC)(TIB)]·3H ₂ O	60	69

H₃BTC = 1,3,5-benzenetricarboxylic acid; btb = 4,4'-bis(1,2,4-triazolyl-1-yl)-biphenyl; H₃nbta = 5-nitro-1,2,3-benzenetricarboxylic acid; H₂oba = 4,4'-oxydibenzoic acid; 4-bpdb = 1,4-bis(4-pyridyl)-2,3-diaza-1,3-butadiene; TIB = 1,3,5-tri(1H-imidazol-1-yl).

investigation is needed to thoroughly understand the adsorption mechanism of CR on the UiO-66-*n*Ti MOFs, the electrostatic attraction seems to be the prevailing factor that determines the adsorption capacity. As shown in Table S3,† the sulfonic acid groups in CR molecules make it a dye with a highly negatively charged surface. Thus it has a strong electrostatic attraction effect with an adsorbent having a positively charged surface. Zeta potential analysis was then carried out to measure the surface charge state of the synthesized samples. The results are summarized in Table S6.† The parent UiO-66 shows a negative zeta potential of -10.2 mV, indicating a negatively charged surface. Thus the electrostatic repulsion with anionic CR molecules explains its low adsorption capacity for CR. Notably, the zeta potentials became positive after the Ti⁴⁺ doping, illustrating the positively charged surfaces of the UiO-66-*n*Ti MOFs. Consequently, negatively charged CR molecules can easily absorb on the positively charged UiO-66-*n*Ti MOF surfaces due to the electrostatic attraction.^{1,4,35,70} Additionally, the zeta potential increases dramatically from 4.8 to 57.1 mV along with the concentration of Ti⁴⁺ for UiO-66-0.7Ti, UiO-66-1.4Ti and UiO-66-2.1Ti. With a further increase in the Ti⁴⁺ concentration, the zeta potentials of UiO-66-2.7Ti and UiO-66-4.0Ti increase only from 57.1 to 60 mV, implying that the aqueous suspensions seem to be steady. The increase of the zeta potential means the increase of the amount of positive charge on the adsorbent surface, which directly leads to the promotion of the adsorption of CR.

Therefore, the efficient removal of CR on the UiO-66-*n*Ti MOFs can be mainly attributed to the strong electrostatic attraction. The Zeta potentials of the samples can be also employed to illustrate the adsorption selectivity shown in Fig. 7. The electrostatic repulsion between cationic dyes MO and RhB and the positively charged UiO-66-2.7Ti results in the relatively low adsorption capacities (Fig. 7b and d). Although MO is also an anionic dye, the fewer negative sulfonic acid groups in the structure is probably the reason for its low adsorption efficiency (Fig. 7c).

The increase of the zeta potentials caused by the incorporation of Ti⁴⁺ in the UiO-66 framework can be explained by the

following speculation. Firstly, due to the different coordination number of Ti⁴⁺ and Zr⁴⁺, the substitution of Zr⁴⁺ by Ti⁴⁺ could result in a decrease in the amount of the hydroxyl groups within the framework. Furthermore, during the adsorption of CR, we observed that when the UiO-66-*n*Ti MOFs were added to the CR solution, the red colored dye solution turned to purple, implying the acid environment,⁷¹ whereas adding the parent UiO-66 to the CR solution did not cause the color change. Thus the decrease of hydroxyl groups of the UiO-66-*n*Ti MOFs caused by the Ti⁴⁺ doping could be the reason for the acid environment of UiO-66-*n*Ti/CR solutions. Secondly, as demonstrated by the TGA results, the Ti⁴⁺ doped samples have more negatively charged deprotonated H₂BDC missing than the parent UiO-66. All these structural changes could make the UiO-66-*n*Ti MOFs more deficient of the negative charges than the parent UiO-66,^{28,55,72} causing the increased zeta potentials in water.

Besides, the porosity of the MOF adsorbent also plays an important role in the adsorption,^{73,74} which is directly reflected on UiO-66-4.0Ti with the most positive zeta potential of 60 mV but a lower adsorption capacity of CR because of the decrease in the accessible active sites due to the reduced surface area (see Table 2 and Fig. 6). Additionally, CR is a dye with a large molecule size (~20.2 × 10.1 × 10.2 Å). It can only theoretically enter a very small fraction of pores between 15 and 20 Å in the samples (Fig. 6b). Thus the diffusion rate in this case can be considered to be very limited. Therefore, all these results suggest that both the electrostatic attraction between CR and the adsorbent and the porosity of the adsorbent dominate the adsorption of CR on the UiO-66-*n*Ti MOFs, but the former is clearly the primary driving force for the adsorption.

3.2.5. Effect of pH on the adsorption and the recyclability of the adsorbent. Generally, the pH of the dye solution has a significant influence on the adsorption, because pH can affect the dissociation of functional groups on the active sites of the adsorbent as well as the structure of the dye molecule, resulting in the change of the charge state between the adsorbent and dye molecule.^{4,66} Therefore, the effect of the CR solution pH on the adsorption of UiO-66-2.7Ti was also investigated. The pH was adjusted from 2 to 10 using 0.1 M diluted HCl or NaOH solutions. As shown in Fig. 10, at pH = 2, the removal rate of CR is almost 100% whereas the removal rate decreases dramatically to about 23% when increasing the pH from 2 to 10, demonstrating that the acid environment facilitates the adsorption of CR on UiO-66-2.7Ti. Likewise, the very positive zeta potential of UiO-66-2.7Ti also decreases gradually to nearly neutral with the increase in the pH. The positive changes on the UiO-66-2.7Ti surface can be gradually neutralized by the hydroxyl at higher pH, resulting in the decrease of the electrostatic attraction with CR molecules.⁶² Additionally, the crystal structure of UiO-66-2.7Ti after the adsorption at various pH values remains the same, demonstrating its excellent structural stability in both acidic and basic solutions (the XRD patterns of UiO-66-2.7Ti after the adsorption at various pH values are shown in Fig. S6†).

Apart from the adsorption capacity and stability of the adsorbent, the recyclability is also an important factor for com-

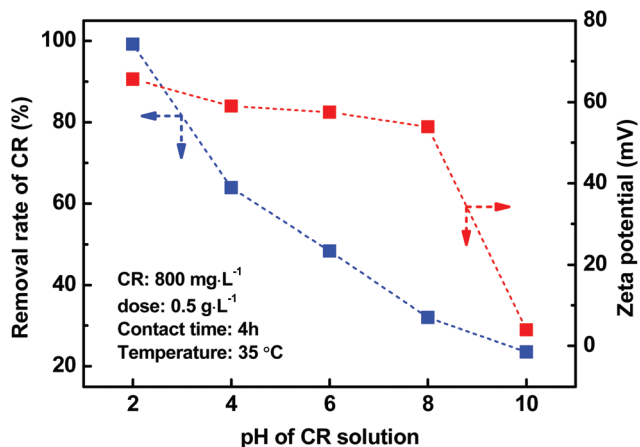


Fig. 10 Effect of pH on the removal rate of UiO-66-2.7Ti (blue curve) and the zeta potential of UiO-66-2.7Ti in aqueous solutions with different pH values (red curve).

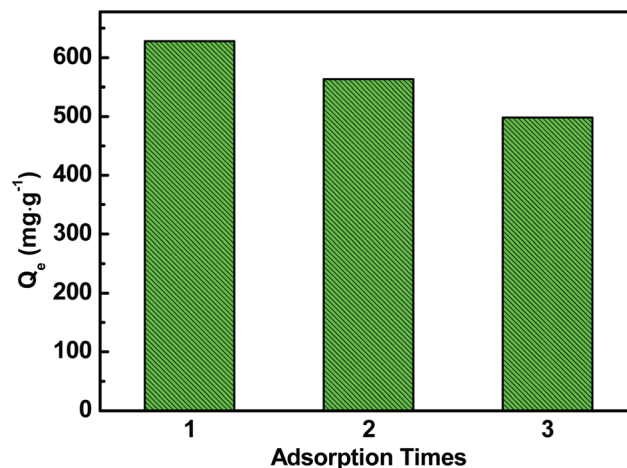


Fig. 12 Recyclability of UiO-66-2.7Ti for the adsorptive removal of CR ($t = 4$ h, $\text{pH} = 7$, $\text{CR} = 400 \text{ mg L}^{-1}$, $\text{doses} = 0.5 \text{ g L}^{-1}$, $T = 35^\circ\text{C}$).

mercial feasibility. The FT-IR spectra of UiO-66-2.7Ti before and after the adsorption of CR are shown in Fig. 11. As can be seen, the main characteristic peaks of UiO-66-2.7Ti remained after the adsorption, which demonstrated its excellent stability. The peak at 1660 cm^{-1} representing the $\text{C}=\text{O}$ asymmetric stretch of the residual DMF disappeared after the adsorption. The shift of the peak from around 1590 cm^{-1} to 1574 cm^{-1} and the appearance of two small peaks at around 1046 cm^{-1} and 824 cm^{-1} are caused by the adsorption of CR molecules.⁷⁵ Additionally, the SEM image of UiO-66-2.7Ti after the adsorption of CR (Fig. S7†) showed that the crystal morphology was maintained after the adsorption. Fig. 12 shows the recyclability of UiO-66-2.7Ti, compared to the first time adsorption, about 86% of the removal rate of CR is maintained after a three time regeneration by ultrasonic-assisted washing with ethanol. Both the mass loss of adsorbent during the regeneration processes and the residual CR molecules on the UiO-66-2.7Ti surface can be the reasons for the reduction of the adsorption capacity.

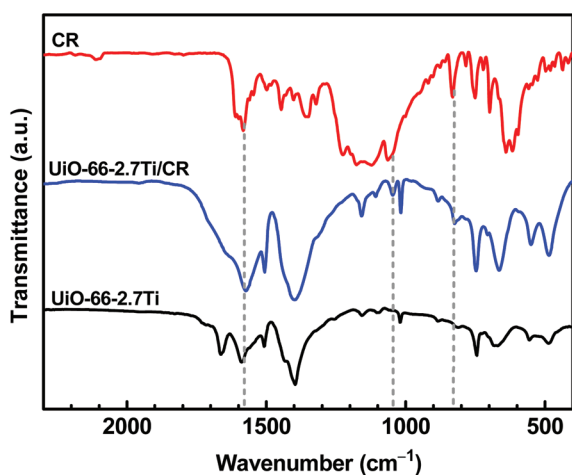


Fig. 11 FT-IR spectra of CR, UiO-66-2.7Ti and CR-adsorbed UiO-66-2.7Ti.

4. Conclusions

In summary, a series of Ti^{4+} doped Zr-based hybrid UiO-66- $n\text{Ti}$ MOFs have been synthesized *via* an *in situ* synthesis method. The Ti^{4+} doping has an obvious influence on the crystal size, morphology as well as the porosity of the samples. Based on the TGA results, it can be seen that the UiO-66- $n\text{Ti}$ MOFs are more structurally defective than the parent UiO-66. The UiO-66- $n\text{Ti}$ MOFs show a highly efficient and selective adsorption ability in removing anionic dye Congo red from water. UiO-66-2.7Ti with 2.72% Ti^{4+} incorporation exhibits the highest adsorption capacity of 979 mg g^{-1} , which is 283 mg g^{-1} for the parent UiO-66. According to the zeta potential measurement, the surface charge state of the samples altered from negative to positive by the incorporation of Ti^{4+} . Therefore, the strong electrostatic attraction between the positively charged surface of UiO-66-2.7Ti and the negatively charged CR molecules greatly promotes the adsorption efficiency. We hope that this work can help to further open the possibility of the synthesis of foreign metal-doped hybrid MOFs directly through the *in situ* synthesis method, and expand the strategies for improving the adsorption performance of MOFs in environmental and energy as well as in other potential applications.

Conflicts of interest

There are no conflicts to declare.

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